

Policy Uncertainty Reduces Green Innovation*

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Abstract

Policy uncertainty can undermine the power of government subsidies to stimulate environmentally friendly research and development. We show that Chinese firms' green R&D falls as the uncertainty of environmental subsidies rises: Exogenous, weather-driven air pollution variability induces subsidies to fluctuate, and firms in areas with high weather-driven subsidy variability undertake less green R&D and hire fewer technical employees, controlling for the average level of subsidies. Heavy emitters and environmental technology firms are more affected. The results also illustrate how policy uncertainty can arise when policymakers are influenced by conditions that are salient but with causes that are difficult to disentangle.

Keywords: R&D, Corporate Investment, Irreversible Investment, Environmental Economics, Policy Uncertainty, Climate Finance

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1. Introduction

Air pollution control efforts are important because airborne particulates have health and economic impacts that are subtle but pervasive. They shorten human lifespans by an average of two years and reduce the healthfulness of the remaining years.¹ In China, air pollution has been estimated to contribute to an astounding 17% of all deaths.² The economic consequences of particulate matter include reduced productivity, human capital formation, economic growth, altered investor behavior and stock prices, and other harms still being uncovered.³

In this paper, we study the effectiveness of environmental subsidies to Chinese firms in terms of the “green” investment that they are intended to stimulate. Such subsidies are intended to overcome the failure to incorporate negative externalities by making projects attractive that would otherwise be unprofitable. When these subsidies become unreliable, firms are presented with an option-to-wait to gather more information about their true persistence. This induces a classic policy uncertainty effect that tends to reduce investment (e.g., Bernanke, 1983; McDonald and Siegel, 1986; Dixit and Pindyck, 1994; Baker et al., 2016; Gulen and Ion, 2016; Engle, 2023). Policy uncertainty is a general concern to firms and can involve tax rates, regulation, trade policy, or other government interventions; the uncertainty in these factors can

¹ <https://aqli.epic.uchicago.edu/pollution-facts/>. For methodological underpinnings, see Chen et al. (2013) and Ebenstein et al. (2017).

² Rohde and Muller (2015).

³ Air quality impacts on economic outcomes include education (e.g., Currie et al., 2009; Zou, 2021), labor supply (Hanna and Oliva, 2015), productivity (Graff Zivin and Neidell, 2012; Chang et al., 2016, 2019; Isen, et al., 2017; He et al., 2020), growth (Ebenstein et al. 2015), happiness (Zheng et al. 2019), neighborhood sorting (Heblich et al., 2021), investor behavior (Li et al. 2021), and financial markets (e.g., Heyes et al. 2016; Huang et al., 2020). See Aguilar-Gomez et al., 2022, for a summary discussion.

be driven by financial crises, election periods (Julio and Yook, 2012) and partisan conflict (Azzimonti, 2018), debt ceiling debates, and so on, and can often be endogenous (Akey et al., 2023). The policy uncertainty literature is justifiably large and discussed in more detail later in the paper. However, there is relatively little research on the effect of policy uncertainty on *green* investment, so its relevance to the burgeoning field of climate and green finance remains to be established.

To fill this gap, we propose a novel behavioral channel to identify the unintended uncertainty of environmental subsidy policies—and we show that such policy uncertainty can discourage the green R&D investment that subsidies are intended to stimulate. Our general approach is as follows. Consider a central planner who periodically allocates subsidies to firms to encourage environmentally friendly investment. The starting point is to figure out where the problems are. It is infeasible to monitor more than a small fraction of the point sources of emissions in a municipal area, let alone thousands of such areas across a large country, so policymakers rely on a limited number of collection devices. With Air Quality Index (AQI) data from many locations in hand, the central planner can establish targets and allocate subsidies to more polluted areas and down to the firms that can make best use of them.⁴

Fig. 1 suggests why this allocation problem is, in practice, far harder than it seems, and it highlights the plausibility of salience as a driver of policy uncertainty. Panel A shows the same

⁴ For more on U.S. and E.U. air quality measurement practices, see <https://www.epa.gov/criteria-air-pollutants/naaqs-designations-process> and <https://www.eea.europa.eu/themes/air/air-quality-management>, respectively. The WHO uses a closely related index which includes most of the same pollutants as the AQI.

building in Beijing on different weekdays in 2014. Panel B connects the pictures to wind speed and the associated AQI. What differentiates the pictures is not the prevailing local emissions, which are stable within a year, but modest differences in wind speed, as wind disperses air pollution. Panel C shows a similar relationship between AQI and rainfall. Rain clears the air by pushing particulates to the ground. Given that weather can both improve pollution and make it worse—importantly, weather varies not just day to day but at lower, policy-relevant frequencies based on average conditions over periods of time—the attribution of pollution becomes difficult, and subsidy allocations are at risk of being affected by weather “noise.”

To further examine how weather-induced policy uncertainty affects green innovation, we develop a simple model that embeds complementary mechanisms, such as wait-and-see motivations under adjustment costs, or simply managerial risk aversion, through which subsidy uncertainty puts a drag on the investment it aims to support. To test the predictions, we assemble a panel of thousands of firms across hundreds of cities in China from 2003 to 2019 (pausing at the Covid onset). For each city-year, we observe weather, air pollution, various city characteristics, and (via translations from Chinese keywords) the environmental subsidies received by local firms, as well as their own “green” R&D investment. We trace through the empirical strategy in which weather fluctuations affect prevailing air quality and potentially induce investment-depressing subsidy variability.

There is robust evidence for this channel. A one-standard-deviation increase in wind and rain variability relative to a city’s long-term average translates into a 13% reduction in green

R&D investment by firms headquartered in the city (controlling for the long-term average subsidy level). More texture comes from examining the cross-sectional effects at the firm level. The firms that respond the most to policy uncertainty include green-tech, manufacturing, and chemicals firms, which is notable to the extent that one might want to incentivize such firms the most. Also, as predicted by the simple model, firms that receive more subsidies and those more likely to be under financial pressure (high debt and/or low interest coverage) are also more subject to policy uncertainty. Finally, in addition to green R&D investment, hiring in environment-related technology positions also appears to be depressed by policy uncertainty. The obvious explanation is that policymakers who allocate the subsidies may be prone to react to immediate, salient circumstances, even though the causes may be, to some extent, beyond the reach of policy.

Documenting the deleterious effect of policy uncertainty on *green* investment, a critical type of investment in light of current and future social impacts and which has not been much studied through the lens of policy uncertainty, is our main contribution.⁵ Our results address this gap. They also give empirical support to surveys that identify regulatory risk as the top climate risk currently facing firms (e.g., Stroebe and Wurgler, 2021). Also, as mentioned, the channel through which this effect likely arises—a response to recent, salient, but “noisy” indicators of a problem susceptible to policy—hints at a new mechanism to generate policy uncertainty that may operate separately from the political shocks that have been extensively studied.

⁵ For example, two recent special issues on climate finance and socially responsible investment contain little discussion of real investment in environmentally friendly technologies.

In addition to air pollution being a major problem in its own right, our mechanism shares an important feature with various other policy settings: It can manifest as an immediate, salient problem that demands a response despite an imperfect understanding of its causes. One concern highlighted in the literature is the endogenous relationship between policy uncertainty and investment (see, e.g., Akey et al. 2023 for a summary). This issue is especially critical in our context, as both subsidies and investments can be influenced by political uncertainty in addition to economic and social considerations. Our approach provides a novel solution to address this issue. In a broader sense, combating air pollution has some similarities to other policy decisions that must react to recent shocks whose nature is not fully clear in the moment, such as climate policy decisions in the absence of a sharp breakdown between manmade and rapidly fluctuating natural sources of temperature rise; or, financial market regulation in the throes of complex meltdown. The exposition and demonstration of this general, potentially behaviorally induced channel for policy uncertainty, as opposed to, e.g., straightforward political shocks, is our more conceptual contribution, applicable to both climate finance concerns and beyond.

Section 2 gives institutional background and outlines simple predictions. Section 3 describes the data. Section 4 documents that policy depends on AQI, which in turn depends on weather, and then combines weather variability and potential underpinnings to motivate an instrument for policy uncertainty. Section 5 shows that policy uncertainty discourages green R&D by firms at the city level and at the firm level, effects which are driven by firms most

affected by the subsidies. It also depresses related technical employment. Section 6 concludes.

An online appendix with background and robustness tests is available.

2. Institutional setting and a simple model

2.1. Environmental subsidies in China

Subsidies to encourage development in desired directions are widely used in China.

Companies that receive subsidies earmarked for environmental purposes may also receive other subsidies, such as those to support agriculture, health, manufacturing, insurance, or tax subsidies that cannot be connected to a specific goal. See Branstetter et al. (2023) for a general discussion of economic subsidies and their relationships to firm productivity and employment.

Environmental subsidies typically take two forms: to cover investment costs or to subsidize the price of final outputs. An example of the first type is a policy announced by the local government of Shenzhen in 2012 to subsidize the development of new technologies related to the circular economy, energy conservation, and emission reduction.⁶ The government directly funded 20%-30% of total investment based on specific criteria, with a maximum subsidy of 15 million RMB per project. Subsidies of this type directly reduce R&D costs, thereby contributing to the profitability of the associated project.

The second type of subsidy is widely used to promote new energy vehicles (NEVs), for example. According to one such policy, NEV producers sell qualified vehicles to customers at a

⁶ https://www.gd.gov.cn/zwgk/wjk/zcfgk/content/post_2530846.html.

subsidized price, with the government reimbursing the firms later.⁷ The qualification of NEVs is based on their pure electric driving range, which depends on research and innovation. Although subsidies of this type do not directly reduce firm costs, they promote sales, which benefits firms as production based on the new technologies achieves economies of scale.

To understand the allocation of these subsidies, one must understand the mechanics of the process. The Chinese government makes and implements environmental policies in a three-step top-down system as institutionalized in the Environmental Protection Law: from the central government to provinces, from provinces to lower governments such as cities, and finally to firms (the online appendix provides more details). The typical process is as follows. The central government moves first by stating its environmental targets either in its regular five-year plans or interim special-purpose regulations. Significantly, the central government often explicitly specifies air quality targets, be they national or at the province or city level, in terms of Air Quality Index (AQI) metrics. After the central government releases goals, provincial and lower governments launch their own five-year plans and local policies, the implementation of which often involves environmental subsidies. The Environmental Protection Law authorizes a high-level government (and its branches) to oversee its lower-level governments (and its branches)—e.g., to monitor and, when necessary, punish relevant officials. Tracking this process, Zhang, Chen, and Guo (2018) document the role of central supervision of local enforcement, while

⁷ https://www.gov.cn/xinwen/2015-04/29/content_2855040.htm.

Chen, Li, and Lu (2018) study the efficacy of target-based environmental performance evaluation systems for local bureaucrats.

For our study, an important point is that the process produces significant uncertainty in subsidy allocations. For example, after the State Council released the national 11th five-year plan in 2007 proposing various targets, the Hebei provincial government released its five-year plan in 2008, targeting the number of days with “good” air quality (an AQI below 100) in its main cities to exceed 80% by 2010. The province’s 12th five-year plan (covering the period 2011-2015) moved the targeting number to 85% days by 2015. After experiencing the heavy pollution in 2013 that had prevailed in northern China, it reset the target in its 13th five-year plan (2016-2020) to achieve 63% days with good air quality in 2020. Each of these changes had implications for subsidies allocated to flow to firms.

“Five-year plans” create uncertainty by definition, but the Chinese government is known for introducing additional uncertainty through interim policy shifts. One example is the “Coal-to-Gas” energy policy described in the online appendix. China wished to replace coal with cleaner natural gas but had to reverse course due to the gas shortage the policy created. Both the adoption and suspension of the policy reinforced uncertainty and illustrated to firms—such as steel manufacturers, which saw production costs double under gas—that it may be privately optimal to delay costly responses to a new policy.

Our firm-level data, described in the next section, show the considerable uncertainty that such policies can introduce to the flow of subsidies to firms. For example, Shandong Lukang

Pharmaceutical Co., Ltd., received 3.8 million RMB of environmental subsidies in 2009 with labels of “ecological compensation fund,” “ecological subsidy fund,” “special funds for industrial pollution control,” and “environmental protection equipment.” This total then varied as follows: 5.1 million RMB in 2010; 1.5 million RMB in 2011; 1.1 million RMB in 2012; 0.2 million RMB or less in 2013-2016; 0.6 million RMB in 2017; and 2.1 million RMB in 2018. Zijin Mining Group Co., Ltd., received subsidies with a dozen environment-related labels over the same period, with totals of 2.9 million RMB in 2008 and 2009; 8.1 million RMB in 2010; 39.3 million RMB in 2011; 22.0 million RMB in 2012; 13.5 million RMB in 2013; 15.7 million RMB in 2014; and 9 million RMB, or less, in 2015-2017.

To summarize, China is a rich setting for our purpose. It resides high on the list of countries with air pollution problems, and it operates a policy environment that involves significant uncertainty in subsidy support. China is not qualitatively unusual, however. Almost every political and regulatory regime presents firms with policy uncertainty that they need to factor into investment decisions.

2.2. A simple model and predictions

Uncertainty, including the part created by weather “noise,” can discourage long-term investments through a number of mechanisms. The classic and most-studied channel is irreversibility (Bernanke, 1983; McDonald and Siegel, 1986; Dixit and Pindyck, 1994), which induces a wait-and-see approach to investment when payoffs are uncertain; empirical support includes Julio and Yook (2012) and Gulen and Ion (2016). Our context is environmental R&D

investments that are supported by uncertain subsidy payments. These green R&D projects may be highly firm-specific and even less marketable than other forms of intellectual property, let alone brick-and-mortar investment, so the irreversibility is potentially extreme. A second mechanism where uncertainty deters investment, which can work in parallel, is managerial risk aversion as in Papanikolaou and Panousi (2012). R&D investments are of course highly risky.

Here we outline a simple model to illustrate how subsidies typically enter the objective function and to derive predictions about how uncertainty affects the subsidized technology. Consider a firm that raises capital to invest in two technologies at the beginning of a period, each of which delivers random payoffs at the end of the period. The first, G , is an early-stage environmental technology that requires green R&D investment and receives an uncertain subsidy S . Following a common structure of subsidies in China (see the online appendix for examples), the subsidy simply enters as an additional payoff to the relevant investment, meaning that the total per-dollar gross payoff of this investment is $\tilde{R}_G \sim N(\bar{R}_G + \bar{S}, \sigma_G^2 + \sigma_S^2)$. The second technology, X , is a traditional R&D technology, or simply capital expenditure, with a per-dollar gross payoff of $\tilde{R}_X \sim N(\bar{R}_X, \sigma_X^2)$. The two payoffs are independent.

The firm borrows all necessary capital at a gross cost C per dollar. This is a stylized notion of financial constraints or pressures, which could increase when the firm's overall leverage is high due to other operations. We assume both investments are profitable on average: $\bar{R}_G + \bar{S} - C > 0$ and $\bar{R}_X - C > 0$. Also assume that $\bar{R}_G + \bar{S} > \bar{R}_X > \bar{R}_G$. Traditional projects have a higher expected payoff than green R&D without the subsidy, thus the role for subsidies.

The firm determines optimal dollar amounts to raise and invest in G and X using mean-variance utility:

$$\text{Max}_{G,X} G(\bar{R}_G + \bar{S} - C) - \frac{\gamma}{2} G^2(\sigma_G^2 + \sigma_S^2) + X(\bar{R}_X - C) - \frac{\gamma}{2} X^2 \sigma_X^2. \quad (1)$$

The firm may derive value from other operations, but we focus only on the part related to the two technologies. The parameter γ denotes the risk aversion that describes how the firm penalizes uncertainty. It is a stand-in for effects that induce the same qualitative behavior, such as irreversibility and adjustment costs or managerial risk aversion. At the cost of complexity, these could be modeled directly. For example, one could allow for differential adjustment costs for the subsidized versus traditional technology. As mentioned above, highly firm-specific green R&D investments are essentially sunk, and differentially higher adjustment costs would simply reinforce the effects below.

In the mean-variance framework with independent risks, optimal investment in a technology is simply its expected return divided by its penalty-adjusted variance:

$$\begin{aligned} G^* &= \frac{\bar{R}_G + \bar{S} - C}{\gamma(\sigma_G^2 + \sigma_S^2)} \\ X^* &= \frac{\bar{R}_X - C}{\gamma\sigma_X^2} \end{aligned} \quad (2)$$

(Eq. (2) is where one can see that a differentially higher penalty for adjusting G would have an especially strong effect.) The ratio of green to traditional investment, which streamlines the analysis of the relative change in the two types of investment, is:

$$\frac{G^*}{X^*} = \frac{\bar{R}_G + \bar{S} - C}{\bar{R}_X - C} \times \frac{\sigma_X^2}{\sigma_G^2 + \sigma_S^2} \quad (3)$$

See the online appendix for further exposition and proofs of the following comparative statics: (1) Higher subsidy policy uncertainty σ_S^2 reduces both the level and share of green R&D. (2) Higher average subsidies $\bar{S}$ increase the level and share of green R&D. We can interpret higher subsidies as higher “subsidy reliance” since the purpose of the subsidy is to address the market failure that causes a gap between the unsubsidized expected return on G and the expected return on X . Higher subsidy reliance also amplifies the impact of policy uncertainty on the share of green R&D. (3) Higher borrowing costs C (informally, more financial pressures) increase the share of green R&D despite the decline in both green and traditional investment. They also intensify the impact of subsidy policy uncertainty on the share of green R&D.

This is not a structural model to fit the full cross-section of Chinese firms. There are other potential subsidy structures, and some are more dynamic, with an interplay between periodic investments and subsidy support. We model a simple case that provides testable qualitative predictions: whether subsidy policy uncertainty reduces green R&D overall, and more so among firms with higher subsidy reliance and measures of high borrowing. In addition, as a falsification (placebo) test, we also test whether these same factors affect traditional capex and R&D, where the theory predicts weaker or no effects.

3. Data

We gather data on firms and their investment, government subsidies, pollution, weather, and city-level characteristics. Appendix A provides a comprehensive list of variables used in

subsequent sections, and Table 1 shows summary statistics. We overview the data here and introduce more detailed measures as the analysis proceeds.

The most novel data are hand-collected green R&D expenditure and environmental government subsidies as reported within the footnotes of the financial statements in the China Stock Market & Accounting Research Database (CSMAR), which we translated from the Chinese. We identify the green component of both R&D efforts and government subsidies from environment-related keywords appearing in the item description, and we manually read to confirm that the subsidies have a clear connection to the environment.⁸

We aggregate firm environmental R&D and granted government subsidies at the city-year level based on the location of firm headquarters. Following prior literature (e.g., Jaffe, 1988; Adams, 1993; Bloom et al. 2002; Adams et al. 2003), we use the log of the RMB value of city-level environment-related R&D, summed across firms, as our measure of *Green R&D*. The annual average Green R&D is 486.5 million RMB from 2009 to 2019. We use the log of the RMB value of the government environmental subsidy in each city-year as our main policy measure, *Green Subsidy*. The annual average raw subsidy is 51.5 million RMB from 2009 to 2019. We exclude the most recent data due to disruptions caused by the Covid pandemic.⁹ We

⁸ The list of environment-related keywords by declining frequency of usage: energy conservation, environmental protection, waste, furnace, pollution, environment, energy, emission reduction, cyclic utilization, cleansing, sewage, desulfuration, waste water, green, recycle, resource conservation, ecologic, water saving, solar power, flue gas, waste heat, dedusting, denitration, pollution discharge, emission, wind electricity, garbage, tail gas, purify, harmless, energy efficiency, low carbon, renewable, air, afforest, electricity saving, sintering machine, clean, high-efficiency motors, lithium iron phosphate, blue sky, electric bus, energy dissipation, coal to gas, exhaust gas emissions, carbonic oxide. These English words are translated from an original list of keywords in Chinese tabulated in the online appendix. The main results do not depend on this precise list of keywords. See Branstetter et al. (2023) for a neural-network-based classification approach for other categories of subsidies.

⁹ We find stronger effects toward the end of our sample.

also collect government expenditures on another government-guided environmental program, industrial *Waste Gas Treatment*, again in log RMB, from EPSnet.¹⁰ This is distinct from subsidies to firms, so we are able to use it to independently confirm policymakers' attention to AQI.

Daily AQI readings are from the Ministry of Environmental Protection of China (MEPC) for major cities in China.¹¹ The annual AQI is the average daily AQI of a city-year. The average annual AQI across cities is 78. By U.S. standards, a day at this level is of "Moderate" concern, whereas Chinese authorities label it "Good." An AQI above 100 is called "Unhealthy for sensitive groups" in the U.S. and "Lightly polluted" in China. A year with such an average pollution level would have seen many individual days with hazardous levels.

The China Meteorological Administration (CMA) provides weather data. We build our instruments for weather fluctuations from two yearly measures. *Windy Days* is the number of days in a year in which wind speed exceeds 5.5 m/s (equivalent to a wind speed of Level 4 or above in the Beaufort Scale). *Rainy Days* is the number of days having more than 25mm of rain (corresponding to Level 3 or above out of the six rain levels categorized by the CMA). *SD Windy*

¹⁰ EPSnet collects city-level waste gas treatment data from China Statistical Yearbook on Environment, National Environmental Statistics Bulletin, and China Environment Yearbook. A snapshot of the province-level investment completed in the treatment of industrial pollution, including waste gas and other types of industrial pollution, for 2014 is at <https://www.stats.gov.cn/sj/ndsj/2014/zk/html/Z0838e.htm>. The funds are provided largely by the state. These data cover 119 cities from 2003 to 2017.

¹¹ The AQI incorporates the concentrations of six air pollutants—sulfur dioxide, nitrogen dioxide, particulate matter 10 microns or smaller (PM10), PM2.5, carbonic oxide, and ozone—based on data from local monitoring stations. Before 2013, China used an Air Pollution Index (API) that covered the first three of these pollutants. As detailed in the online appendix, the two indices are highly correlated and used comparable numerical reference points for policymakers. Hence, we use API data before 2013 but for simplicity refer to the whole series as AQI.

Days and *SD Rainy Days* are the standard deviations of *Windy Days* and *Rainy Days* in a city over the prior six years. Importantly, this varies substantially across cities.

We also control for city and firm characteristics that may affect subsidies and firm behavior for other reasons. Our main city-level control variables come from CNRDS (which digitalizes data from China City Statistical Yearbook), including log consumption, log GDP, GDP growth, and population growth. Consumption data are available only at the province level, but this does not affect our inferences because of city fixed effects.

More firm-level characteristics come from Wind, CSMAR, and the Chinese Research Data Services Platform (CNRDS). These describe the capital structure, profitability, and investment opportunities of firms, which could also affect firm decisions (e.g., Jaffe, 1988; Adams, 1993; Minton and Schrand 1999; Bloom et al., 2002; Adams et al., 2003). For our city-level analysis, we weight by firm assets to obtain the city-level control variables in Table 1. Our main specifications control for city or firm effects in city-level and firm-level analysis, respectively, so it is not surprising that the inclusion of firm characteristics, the exclusion of financial firms, or other such changes have little empirical impact.

The resulting sample includes an unbalanced panel of 3,168 listed firms, across 352 Chinese cities, from 2003 to 2019. As mentioned above, we unfortunately must exclude the most recent data due to the impact of the Covid pandemic, because several years of contiguous and consistent data are needed to construct policy volatility at the city-year level.

4. Pollution and subsidy levels

We start with a cross-city investigation of the determinants of green subsidies in a given year. The determinants include both weather and other characteristics of the city and firms within the city. We regress the sum of green subsidies reported by firms in the city in a Fama-MacBeth (1973) specification, by which coefficients are averaged across cross-sections to yield standard errors robust to cross-sectional correlation:

$$\text{Green Subsidy}_{j,t} = a_0 + a_1 \times AQI_{j,t-1} + A \times X_{j,t-1} + \varepsilon_{j,t}, \quad (4)$$

where $\text{Green Subsidy}_{j,t}$ denotes the log of the total environmental subsidy provided to firms in city j in year t , $AQI_{j,t-1}$ is the average Air Quality Index in the previous year, and the vector $X_{j,t-1}$ is a set of control variables including city characteristics and asset-weighted averages of the characteristics of firms in city j . City characteristics are log consumption, log GDP, GDP growth rate, and population growth rate. Firm characteristics averaged at the city level are asset turnover, return on assets, leverage, firm size, cash holdings, capital expenditures, profit margin, and lagged stock return. In light of the yearly variation in subsidies granted to the firms in the examples above, annual is a reasonable frequency to study.

The results are in Table 2. The first model examines the effect of city characteristics only. The takeaway is that wealthier cities receive considerably more environmental subsidies, consistent with the univariate correlation from the summary statistics. The coefficient indicates that a 1% increase in GDP is associated with a 2.97% increase in green subsidy. Several political

and economic processes might lead to this intuitive relationship; we simply treat GDP as a control variable.

Our interest centers on a_1 . The next model shows that year t environmental subsidies are, on average, higher in cities with worse air pollution in year $t-1$ (higher AQI). When we control for city characteristics in Model (2), a one-standard-deviation greater AQI is associated with a 0.17 standard deviation greater green subsidy in the following year.¹² Other models in Table 2 show that these conclusions are largely unaffected by the inclusion of firm characteristics, which for brevity we do not report. In the online appendix, we show that the policy-AQI relationship is robust in the post-2013 sample and to variations such as using the city's AQI rank rather than its level.

Since top-down targets are explicitly based on AQI, as discussed earlier, the fact that subsidies depend on recent AQI is not surprising. The bureaucratic concern with the air quality figures themselves is further evident in the manipulation that occurs to keep figures just below numerical targets and round numbers (e.g., Andrews, 2008). As a final confirmation of the importance of this pollution measure, we study another government-guided environmental program, city-level expenditures on treatment of industrial waste gas. With *Waste Gas Treatment_{j,t}* as the dependent variable, we find qualitatively similar results. In Model (5), a one-standard-deviation increase in AQI gives rise to a 0.13 standard deviation

¹² $(\hat{a}_1 \times SD\ AQI)/SD\ Subsidy = (0.05 \times 25.3)/7.4 = 0.17$. (The standard deviations relevant to Table 2 are based on additional observations than those in Table 1, which summarizes the primary regression sample.)

increase in the log of waste gas treatment.¹³ Further specifications are reported in the online appendix.

Finally, in an investigation omitted for brevity, we conducted a textual analysis of city-level government reports to further document the effect of recent conditions on policy. We found that worse AQI readings in the prior year predict an increase environmental keywords, and fewer economic keywords, in the local government's annual government report. This again ties subsidies and local bureaucratic attention to recent environmental conditions.

In summary, subsidy expenditures meant to combat air pollution are directed toward the most-polluted areas, as measured by AQI. This is intuitive and sensible and consistent with how air quality targets are communicated in policy circles.

5. Effect of subsidy policy uncertainty on green R&D and employment

This section contains our central empirical results. We start by develop the connections between weather, AQI, and subsidy policy into a weather-based instrument for subsidy policy uncertainty. We then examine the effect of this policy uncertainty on green R&D investment.

5.1. Preliminary: the effect of weather on AQI

Although AQI endeavors to measure underlying point sources of pollution, Fig. 1 shows that it is also influenced by wind and rain. In the figure, this variation is at a daily or weekly frequency and for a single city. However, just as some days are windier and rainier than others,

¹³ $(\hat{a}_1 \times SD\ AQI)/SD\ Subsidy = (0.04 \times 25.3)/7.9 = 0.13$.

so are some months or years, frequencies that are more relevant to policymaking. To show this, we relate AQI to weather in a very simple specification:

$$AQI_{j,t,s} = b_0 + b_1 \times Wind\ Speed_{j,t,s} + b_2 \times Rain\ Volume_{j,t,s} + B \times X_{j,t} + \delta_j + \delta_t + \varepsilon_{j,t,s} \quad (5)$$

where $AQI_{j,t,s}$ denotes the average daily AQI of city j in the heating or non-heating season s of year t and $Wind\ Speed_{j,t,s}$ and $Rain\ Volume_{j,t,s}$ are the average daily wind speed and rain volume in the same city and period, respectively. Due to the impact of the coal-intensive heating season on air pollution in northern cities (Almond et al., 2009; Chen et al., 2013; Ebenstein et al., 2017), we confirm the AQI-weather relationship separately in heating and non-heating seasons. $X_{j,t}$ includes city characteristics as before, and the specifications also include city and year fixed effects with double-clustered standard errors.

A scientific air pollution model would consider humidity, temperature, wind direction relative to nearby cities, cloudiness, humidity, local geography, and other weather effects on air quality (Cai et al., 2017; Zhang et al., 2018b). Eq. (5) simply reflects how wind reduces the accumulation of pollutants near ground level and rain pushes down particulates and washes them away. These first-order effects are strong, so we do not need a complex pollution model.

The results in Table 3 confirm expected relationships between the AQI and the weather. This is true for heating and non-heating seasons and for northern cities separately. In the sample of all cities (northern cities), for example, an increase of the average wind speed by one meter per second is associated with a decline of heating-season AQI of 3.15 (5.13) points. Thus, we

easily confirm a scientific preliminary: weather has a robust, causal effect on city-year AQI, controlling for city and year fixed effects. See the online appendix for more specifications.

5.2. *First stage: weather as an instrument for policy uncertainty*

As often discussed in the policy uncertainty literature, any mechanism in which investment is negatively related to economic uncertainty, and in which policy itself responds to uncertainty, can lead to an endogenous connection between policy uncertainty and investment. Akey et al. (2023) summarize several such mechanisms, which necessitate the identification based on exogenous sources of policy uncertainty. Researchers have frequently focused on political shocks. Pastor and Veronesi (2012, 2013) study consequences for asset prices, while Baker et al. (2014) emphasize trends in government activity and political polarization. Gowrisankaran et al. (2025) explore the implications of legal uncertainty in the context of air quality standards. Policy uncertainty measures in Baker et al. (2016) are built around fiscal and monetary policy, regulation, and national security.

In our context, the previous results suggest considering weather as the basis for instrument for subsidy policy uncertainty. As a more intuitive motivation for this approach, consider the following anecdote. On November 19, 2010, the Air Quality Index (AQI) measured at the U.S. Embassy in Beijing reached 562. The cheeky programmer who set up the Embassy’s automatic Twitter report apparently never imagined that air pollution would reach the level of a forest fire, and air quality was broadcast as “crazy bad.” “An official said there was insufficient research to explain why the pollution haze has been so murky this week. The culprit is likely to

be a combination of weather, traffic and coal-burning power stations.”¹⁴ Although the cause of the pollution spike could not be pinned down, our data show that environment-related subsidies to Beijing-headquartered firms increased by 78% the next year.

Conceptually, a weather-based instrument for subsidies could arise from a combination of realistic political pressures and a set of highly intuitive biases that are well-known in individual choice settings. In China, the bureaucrats who make regional and city-level subsidy allocation decisions operate in an institutional context with their own incentives and possess authority and discretion that may exceed their scientific qualifications. They, and the populations they serve, may be subject to such everyday biases as salience, extrapolation, projection, recency, and availability. Such “recency biases” tend to predict a strong response to extreme, recent pollution readings that, potentially, exceeds the fundamental basis for regulation.

In particular, attention and decision making can be guided by *salient* phenomena such as recent, unusually high air pollution levels; slightly weaker wind or lower precipitation, although they have large effects on observed pollution, are much less salient. One might also *project* the efficacy of high subsidies from beliefs based on *extrapolating* present conditions, again not fully accounting for the subtle role of recent weather patterns. Several studies document a bias toward overweighting *recent* conditions (again, prevailing pollution levels) per se. These biases are interrelated and overlapping, e.g., both salience and recency can drive an *availability* bias.¹⁵ Representativeness, reinforced by vividness, is yet another related pattern.

¹⁴ <https://www.theguardian.com/environment/blog/2010/nov/19/crazy-bad-beijing-air-pollution>.

¹⁵ See, for example, Taylor and Thompson (1982) and Bordalo et al. (2022) (and references therein) on salience;

Some of these biases have been discussed in the context of air pollution by Chang, Huang, and Wang (2018), who find that demand for health insurance fluctuates with air pollution. As Chang et al. (2018) and Busse et al. (2015) point out, the biases generate similar predictions in real data, so we consider them a related group that drive a novel mechanism that transfers uncertainty in weather into uncertainty in policy. We also note that in a governmental system that prioritizes stability and articulates quantitative targets, Chinese bureaucrats may face political or institutional imperatives to simply “do something” about high pollution. When such pressure is driven not by underlying emissions but partly by an improper attribution of weather, it can also be seen as deriving from recency biases.

With such motivations in mind, we define policy uncertainty $PU_{j,t-5:t}$ as the standard deviation of the characteristics-adjusted subsidy in city j over the prior six-year period. Measuring uncertainty based on time-series volatility in policy is natural and similar to Segal et al. (2015), Fernandez-Villaverde et al. (2015), and others. The six-year rolling window follows Minton and Schrand (1999) and trades off sample size and older, less relevant data. A city’s characteristics-adjusted subsidy is computed from the residuals of Model (1) of Table 2. We adjust for city characteristics as they can influence the routine resource allocation by the central government, e.g., wealthier or faster-growing cities may be associated with more pollution and deserve additional subsidy allocation and/or authorities may be more concerned with pollution in

Loewenstein et al. (2003), Conlin et al. (2007), and Busse et al. (2015) on projection bias; Greenwood and Shleifer (2014) on extrapolation; Hogarth and Einhorn (1992) on recency; and Tversky and Kahneman (1973) on availability and representativeness.

such cities for political reasons. Results in the online appendix indicate that the exact window size or various details of the characteristics-adjustment approach are unimportant.

We begin with simple OLS results in Table 4 for reference. The *PU* coefficient is hard to interpret due to endogeneity, but it is at least consistent with a negative correlation between policy uncertainty and green R&D investment. The first-stage regression in our IV isolates the component of this policy uncertainty due to recent weather variability:

$$PU_{j,t-5:t} = c_0 + c_1 \times SD \text{ Windy Days}_{j,t-6:t-1} + c_2 \times SD \text{ Rainy Days}_{j,t-6:t-1} + C \times X_{j,t} + \delta_j + \delta_t + \varepsilon_{j,t}. \quad (6)$$

The weather variability terms are the standard deviation of the number of windy days and rainy days per year from year $t - 6$ through $t - 1$. The one-year lag between policy uncertainty and weather fluctuation follows the timing convention of Eqs. (4) and (5), in which weather conditions affect contemporaneous AQI and next year's policy. One can imagine slower, faster, or heterogeneous responses; robustness tests indicate that this time convention is not crucial. The specifications again control for city characteristics and city and year fixed effects.

The first-stage IV results are in Model (2) of Table 4. As hypothesized, unusually variable weather is associated with unusually variable policy relative to the city's norm. Specifically, a one-standard-deviation increase in wind and rain fluctuations versus local norms leads to a 0.228-standard-deviation increase in *PU*.¹⁶ The strength of the weather variables, with an F-

¹⁶ $(\hat{c}_1 \times SD(SD \text{ Windy Days}) + \hat{c}_2 \times SD(SD \text{ Rainy Days}))/SD \text{ PU} = (0.01 \times 25.24 + 0.16 \times 1.96)/2.48 = 0.228$.

statistic of 14.5, supports a weather-based component of policy uncertainty to use as an instrument. Ultimately, we construct:

$$\widehat{PU}_{j,t-5:t} = \hat{c}_1 \times SD \text{ Windy Days}_{j,t-6:t-1} + \hat{c}_2 \times SD \text{ Rainy Days}_{j,t-6:t-1}. \quad (7)$$

as an instrumental variable.

5.3. Second stage: policy uncertainty and green R&D

We proceed with the second-stage regression that links green investment to the exogenous component of policy uncertainty. The first outcome variable is green R&D spending by firms across city j , which is a natural unit of aggregation given the city-level variation in weather. We estimate:

$$\begin{aligned} \text{Green R\&D}_{j,t+1} \\ = d_0 + d_1 \times \widehat{PU}_{j,t-5:t} + d_2 \times \text{Avg Subsidy}_{j,t:t-5} + D \times X_{j,t} + \delta_j + \delta_t + \eta_{j,t+1} \end{aligned} \quad (8)$$

where $\text{Green R\&D}_{j,t+1}$ is the log of the aggregate environmental R&D investments that firms of the city make in the following year. The key hypothesis is whether failing to endogenize weather-caused air quality variation gives rise to exogenous policy uncertainty, which in turn diminishes firms' incentives to invest in green technology. This would be reflected in a negative coefficient on the instrumental variable, the basic prediction. We also include the average recent subsidy for the city since a higher R&D subsidy may associate with higher green R&D.

The remaining columns of Table 4 tabulate the second-stage results. The subsidy level itself is indeed consistent with a stimulative effect on green R&D. A one-standard-deviation increase in

the subsidy is associated with a 0.158-standard-deviation increase in city-level green R&D.¹⁷

See Howell (2017) for U.S. evidence that environmental subsidies boost innovation.

Policy uncertainty, however, exerts an opposing influence. For instance, Model (5), viewed in conjunction with Model (2), suggests that a one-standard-deviation increase in wind and rain fluctuations translates into a 0.132 standard deviation reduction in R&D.¹⁸ A direct comparison between the magnitudes of the positive level effect and the negative uncertainty effect suggests that the consequences of policy uncertainty are material, with subsidy uncertainty meaningfully undermining the effect of the subsidy on the desired corporate activity.

It is worth mentioning that the value of the IV approach here becomes clear when the second-stage coefficient is compared to the smaller OLS coefficient. The conventional reasons for this pattern include one or more of the following: an omitted variable; measurement error that arise because realized policy uncertainty is noisy, subject to extra variation that does not relate to green R&D investment; and/or, the fact that the IV estimate describes the effect on firms for which green subsidy policy uncertainty really matters for green R&D, as opposed to the average treatment effect over the entire sample of firms from many industries. This last mechanism, which brings to mind the comparative statics and whose empirical relevance is easy to envision, is supported further in the firm-level heterogeneity below.

5.4. *Robustness checks*

¹⁷ $(\hat{d}_2 \times SD \text{ Avg Subsidy})/SD \text{ Green R\&D} = (0.34 \times 4.44)/9.54 = 0.158.$

¹⁸ $(\hat{c}_1 \times SD (SD \text{ Windy Days}) + \hat{c}_2 \times SD (SD \text{ Rainy Days})) \times \hat{d}_1/SD \text{ Green R\&D} = (0.01 \times 25.3 + 0.11 \times 1.95) \times (-2.23)/9.54 = -0.132.$

These patterns are robust to alternative specifications and variable construction choices and pass placebo tests. We report the results of some variations in Table 5 and additional specifications in the online appendix.

A minority of cities report no green subsidies at all in a given year, so Model (1) includes only city-year observations with nonzero R&D. The results are little changed. In the appendix, we report similar findings when excluding observations with zero subsidies in general or excluding any city's observations before its first nonzero subsidy. Model (2) varies lags and timing. We assume a one-year lag as an approximation to how observed AQI may affect the allocation of subsidies, but a year t allocation could be determined in the late part of year $t-1$, when the year $t-1$ average AQI is not fully realized. Or some cities may allocate subsidies over multiple years. In any event, the results are similar upon averaging AQI over two years.

In Model (3), we control for the variation in non-weather related AQI. We regress pollution on weather conditions and treat the residuals as non-weather-related pollution variation. Controlling for the time-series volatility of these residuals has no impact. This further suggests a unique role of weather fluctuations on policy uncertainty. In Model (4), we base routine allocations on both city and aggregated firm characteristics as opposed to only city "macro"-level control variables. The results are unchanged. In Model (5), we compare the second half of the sample. The point estimate turns out to be stronger in the second half of the sample. This may reflect growing concern for the environment and the use of subsidies, along with their variability, as environmental policy tools.

The remaining Models (6)-(8) contain important placebo tests that point to a specific connection between environmental policy uncertainty and, specifically, green R&D as opposed to other components of investment. First, weather-induced green policy uncertainty does *not* affect *non-green* R&D. Second, it does not affect city-year level capital expenditure by sample firms. Third, to the extent that one might be concerned that weather-induced fluctuations in AQI are so severe as to depress real estate investment as a whole, it does not affect city-year level capex by real estate firms controlling for city fixed effects. These results support the specific channel we hypothesize and the exclusion restriction of the IV approach.

The online appendix provides numerous further robustness tests involving the list of environmental keywords, the effects of winsorization, the inclusion of emission variables, the inclusion of controls for the volatility of AQI, and an average annualized regression method (e.g., Minton and Schrand, 1999; Bates et al. 2009) as opposed to a panel specification.

5.5. *Heterogeneous impacts of policy uncertainty*

We now turn to firm-year data. The model's comparative statics suggest two predictions about which firms might be more sensitive to subsidy policy uncertainty—those that are more subsidy reliant and those for whom subsidies would be more helpful for relaxing financial constraints. We use firm-level investment in the second stage and an interaction of the instrument for uncertainty and “firm types”:

$$\begin{aligned} \text{Green R\&D}_{i \in j, t+1} = & e_0 + e_1 \times \widehat{PU}_{j, t-5:t} + e_2 \times \text{Firm Type}_{i \in j, t} \times \widehat{PU}_{j, t-1} \\ & + e_3 \times \text{Avg Subsidy}_{j, t-5:t} + E \times X_{i \in j, t} + \delta_i + \delta_t + \eta_{i \in j, t+1}. \end{aligned} \tag{9}$$

$Green\ R\&D_{i\in j,t+1}$ is the year $t+1$ investment made by firm i of city j . $Firm\ Type_{i\in j,t}$ is an indicator for firm types such as green-tech suppliers, firms in a metropolitan area, large firms, energy firms, manufacturing firms, mining firms, and chemicals firms. The indicator itself is absorbed by firm fixed effects.

The results are in Table 6. The effect of weather-induced policy uncertainty also appears in firm-level specifications. The second model shows that policy uncertainty has a far stronger effect on green-tech suppliers than on the average firm. This is intuitive, but not necessarily desirable, to the extent that environmental R&D subsidies are intended to incentivize such firms in particular.

The next several models explore several characteristics related to the headquarters location of the firm, size (top 10% assets), state ownership, and reliance on environmental subsidies. We find that firms with more reliance on environmental subsidies are more sensitive to environmental policy uncertainty. The latter effect, like the differential effect on green tech firms, is as predicted by our simple model. The former effect might be related to the higher cost of hiring and maintaining R&D employees in these cities (our tests below suggest that uncertainty also reduces R&D staffing). Size and state ownership status do not mediate the influence of policy uncertainty, on the other hand, in the absence of other interactions. It is possible that the higher government dependence of such firms is offset by an informal insulation from the full variability in subsidies.¹⁹

¹⁹ Table 6 allows sample firms to have different data lengths in our sample. In the online appendix, we apply the same specifications to the subset of firms that exist for the full sample period for the second-stage regression. The results

Some listed firms have a complex geographic footprint, making it unclear which city's policy affects them the most. We create a dummy variable that equals one if a firm is registered in one city and has its headquarters in another, but we find no detectable differential effect of policy uncertainty on these firms.

We next examine the impact of policy uncertainty across industries. The online appendix provides a detailed matching between China's industry classification provided by China Securities Regulatory Commission (CSRC), two-digit SIC codes, and Fama and French's (1997) classifications. Manufacturing and water conservancy, environment, and public facility management are most affected. Manufacturing includes many important polluting industries, such as chemicals and metals, and the public facilities designation includes environmental firms. The last column indicates that these embody independent sensitivities to policy uncertainty.

Table 7 examines some further cross-sectional predictions of the model. Recall that the model's comparative statics included stronger effects on green R&D when the firms were more subsidy reliant, which was confirmed in Table 6, but also when financial constraints were more likely to be binding.

Measuring financial constraints is notoriously challenging, with no consensus in the literature, and the task is made harder still by the differences in operational structures across the full cross-section of listed Chinese firms. We approach this question by deferring to proxies for financial constraints provided within the CSMAR dataset itself. *Over-indebtedness* is CSMAR's

indicate that mature firms, like large firms and SOEs, are also affected by policy uncertainty.

calculation of the actual debt ratio minus the target debt ratio²⁰; *Interest Coverage Ratio A* is CSMAR's calculation of Net Profit plus Income Tax Expense plus Interest Expense, all divided by Interest Expense; and *Interest Coverage Ratio B* excludes Income Tax Expense.

The results are consistent with predictions. Among the firms particularly affected by policy uncertainty—green tech—we find that those with relatively high levels of debt (relative to CSMAR-assessed target) or low interest coverage ratios are more sensitive to policy uncertainty. These effects are also economically significant. For instance, in Model (2) the coefficients for $\widehat{PU} * Green\ Tech$ and its further interaction with *Over-indebtedness* are -0.60 and -0.55, respectively. The first coefficient indicates that a one-standard-deviation increase in wind and rain fluctuations leads to a 5.8% standard deviation reduction in the firm-level green R&D for Green Tech firms. The second one implies that a one-standard-deviation increase in *Over-indebtedness* further enhances this negative impact by 14% ($0.1557 * -0.55 / -0.60 = 14\%$, where 0.1557 is the standard deviation of *Over-indebtedness*).

5.6. Green R&D employment

Bloom (2009) estimates sizeable hiring adjustment costs, which imply the same costly reversibility mechanism and optimal inertia that appears in R&D spending. We can also employ the two-stage analysis to examine the effect of green subsidies on labor input. Returning to the

²⁰ The CSMAR database applies the methodologies of Harford et al. (2009) and Denis and McKeon (2012) to estimate a Tobit regression model to predict the leverage [Total Liabilities/Total Assets] for a Chinese firm. The predicted leverage, which represents the target debt ratio, is restricted to lie between zero and one. To estimate leverage, the model incorporates various lagged firm characteristics, including a dummy variable for identifying State-Owned Enterprises (SOEs), return on assets (ROA), the industry median debt-to-asset ratio, the growth rate of total assets, the proportion of fixed assets to total assets, the natural logarithm of total assets, and the ownership share of the firm's largest shareholder.

city-year level, we define *Green R&D Employment* as the logarithm of one plus the total number of R&D and technical employees in firms in the city that report environmental R&D investments. This is a count of the jobs most likely to be supported by the subsidies in question.

We then repeat the second-stage analysis with this dependent variable. The results are in Table 8. In addition to reducing green R&D, shocks to policy uncertainty also appear to reduce associated employment. A one-standard-deviation increase in wind and rain fluctuations leads to a 0.153 standard deviation reduction in *Green R&D Employment*. While the functional form of R&D output is hard to know, this effect is of the same order of magnitude as the effect on R&D.

6. Conclusion

In this paper, we document that economic policy uncertainty has “real” consequences for green innovation, connecting these two growing literatures. Our specific setting involves Chinese subsidy to support environmental R&D. We find that exogenous variation in the subsidies, generated by the effects of weather on air quality, reduces the amount of green R&D and technical employment that the subsidies are intended to promote. The effects are more pronounced among firms where theory predicts—environmentally-oriented, subsidy reliant, and more financially pressured.

We encourage additional research on the real side of green investment. The paper also points to a new mechanism for policy uncertainty. Policy uncertainty may be created not just by political shocks, as has been well studied, but also by “recency biases” that have been robustly

documented in individual choice settings. In this paper, for example, policymakers appear to respond to a weather-driven component of air pollution, generating additional policy uncertainty. But major economic and financial policy decisions often must be made in settings where the problem is salient and immediate even though the causes are a complex mix of exogenous and endogenous factors. The appropriate policy response may not be clear until the dust settles.

CRedit authorship contribution statement

Mengyu Wang: Writing – original draft. **Jeffrey Wurgler:** Writing – original draft. **Hong Zhang:** Writing – original draft.

Declaration of competing interest

We have no conflicts of interest to disclose

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Appendix A.

Variable definitions and sources

Variables in Panels A through D are at the city-year level. In indicated cases, they are aggregates or asset-weighted averages of the firm-year-level variables in Panel E.

Variable	Definition	Source
A. Policy and Policy Uncertainty		
<i>Green Subsidy</i>	The logarithm of one plus city environmental (“Green”) subsidy (RMB value).	CSMAR
<i>PU</i>	Policy uncertainty in a city-year estimated as the city characteristics-adjusted standard deviation of <i>Green Subsidy</i> in the six-year window $t - 5:t$.	(constructed)
$\widehat{PU}$	Weather-driven policy uncertainty is estimated in two steps. First, Model (2) of Table 4: $PU_{j,t-5:t} = c_0 + c_1 \times SD\ Windy\ Days_{j,t-6:t-1} + c_2 \times SD\ Rainy\ Days_{j,t-6:t-1} + C \times X_{j,t} + \delta_t + \delta_j + \varepsilon_{j,t}$, where X denotes city characteristics. Second, the projection: $\widehat{PU}_{j,t-5:t} = \hat{c}_1 \times SD\ Windy\ Days_{j,t-6:t-1} + \hat{c}_2 \times SD\ Rainy\ Days_{j,t-6:t-1}$.	(constructed)
<i>Waste Gas Treatment</i>	The logarithm of one plus city-level investment in treatment of industrial waste gas (RMB value)	EPSnet
B. Green Investment		
<i>Green R&D</i>	The logarithm of one plus aggregate environmental R&D investment (RMB value) of firms in the city (the aggregate of the firm-level <i>Green R&D</i>).	CSMAR
<i>Green R&D Employment</i>	The logarithm of one plus the aggregate number of R&D and technical employees among firms with nonzero environmental R&D investment in the city.	Resset
C. Air Pollution and Weather		
<i>AQI</i>	The AQI in a city (annual average of daily values unless otherwise indicated).	Ministry of Ecology and Environment
<i>Wind Speed</i>	The daily wind speed in the city in m/s.	China Meteorological Administration
<i>Windy Days</i>	The number of days in the year in which the city’s wind speed is level 4 or above on the Beaufort Scale (i.e., more than 5.5 m/s).	(constructed)
<i>SD Windy Days</i>	The standard deviation of the number of windy days in the city per year in a six-year rolling window.	(constructed)

<i>Rain Volume</i>	The volume of rain in the city in mm.	China Meteorological Administration
<i>Rainy Days</i>	The number of days in the year in which the city's rain volume is level 3 or above (i.e., more than 25mm/day).	(constructed)
<i>SD Rainy Days</i>	The standard deviation of the number of rainy days in the city per year in a six-year rolling window.	(constructed)
D. City Characteristics		
<i>Consumption</i>	The logarithm of consumption (RMB value).	CNRDS
<i>GDP</i>	The logarithm of GDP (RMB value).	CNRDS
<i>GDP Growth</i>	The GDP growth rate.	CNRDS
<i>Population Growth</i>	The population growth rate.	CNRDS
E. Firm Characteristics		
<i>Green R&D</i>	The logarithm of one plus environmental R&D investment (RMB value) of a firm.	CSMAR
<i>Green Tech</i>	One if a firm provides environmental/green products and services else zero. This includes firms that belong to the ecological protection and environmental governance industry (CSRC N77).	CNRDS
<i>Metropolitan</i>	One if a firm is located in Beijing, Shanghai, Guangzhou, or Shenzhen, else zero.	CNRDS
<i>Large</i>	One if firm total assets is in the top decile else zero.	WIND
<i>SOE</i>	One if a firm is state owned else zero.	CNRDS
<i>Subsidy Reliant</i>	One if a firm's ratio of environmental subsidy to total assets is higher than the median else zero.	CSMAR
<i>Different Cities</i>	One if a firm's office address and registered address are in different cities else zero.	CSMAR
<i>Manufacturing</i>	One if a firm belongs to the manufacturing industry (CSRC prefix C) else zero.	CSMAR
<i>Chemical</i>	One if a firm belongs to the chemical industry (CSRC C25, C26, C28, C29, C41, and C43) else zero.	CSMAR
<i>Metal</i>	One if a firm belongs to the metals industry (CSRC C31, C32, C33, C65, and C67) else zero.	CSMAR
<i>Public Facility</i>	One if a firm belongs to the public facility industry (CSRC prefix N) else zero.	CSMAR

<i>Over-Indebtedness</i>	Over-indebtedness is the degree of excess debt that is calculated as the actual debt ratio minus target debt ratio. This is directly from CSMAR.	CSMAR
<i>Interest Coverage Ratio A</i>	Interest Coverage Ratio A equals (Net Profit + Income Tax Expenses + Interest Expense) / Interest Expense. It is scaled by 10,000 in the regression.	CSMAR
<i>Interest Coverage Ratio B</i>	Interest Coverage Ratio B equals (Net Profit + Interest Expense) / Interest Expense. It is scaled by 10,000 in the regression.	CSMAR
<i>Turnover</i>	Total assets turnover defined as revenue divided by total assets.	WIND
<i>ROA</i>	Operating income divided by total assets.	WIND
<i>Leverage</i>	Book value of total debt divided by total assets.	WIND
<i>Size</i>	The logarithm of total assets (RMB value).	WIND
<i>Cash</i>	Cash holdings divided by total assets.	WIND
<i>Capex</i>	Capital expenditure divided by total assets.	WIND
<i>Profit Margin</i>	Net income divided by revenue.	WIND
<i>Stock Return</i>	Prior-year stock return.	CSMAR

Table 1**Summary Statistics**

Variables are defined in Appendix A and are generally at the city-year level in an unbalanced panel of up to 352 cities in the pre-Covid period 2009 to 2019 (the *SD* weather variables, constructing using a six-year rolling window, are defined using weather data back to 2003), which is the sample used in our main analyses of policy uncertainty and green investment.

	N	Mean	SD	5%	25%	50%	75%	95%
Policy								
<i>Green Subsidy</i>	1340	12.67	6.40	0.00	12.61	15.22	16.64	18.49
<i>Waste Gas Treatment</i>	1340	13.96	7.93	0.00	13.48	18.07	19.24	20.32
Green Investment								
<i>Green R&D</i>	1340	11.00	9.54	0.00	0.00	17.03	19.44	21.62
<i>Green R&D Employment</i>	1340	4.00	3.70	0.00	0.00	5.09	7.30	9.16
Weather and Pollution								
<i>AQI</i>	1340	77.79	24.35	47.35	62.18	73.59	87.18	122.41
<i>Wind Speed</i>	1340	4.60	1.12	2.85	3.92	4.55	5.25	6.48
<i>Windy Days</i>	1340	103.03	75.20	4.00	39.50	92.50	151.00	246.00
<i>SD Windy Days</i>	1340	23.73	25.24	2.49	8.86	16.16	28.56	72.63
<i>Rain Volume</i>	1340	2.70	1.60	0.57	1.47	2.34	3.76	5.69
<i>Rainy Days</i>	1340	10.86	7.78	1.00	5.00	9.00	15.50	26.00
<i>SD Rainy Days</i>	1340	3.30	1.96	0.82	2.00	2.94	4.27	7.03
City Characteristics								
<i>Consumption</i>	1340	9.61	0.47	8.72	9.33	9.63	9.94	10.33
<i>GDP</i>	1340	10.63	0.78	9.36	10.15	10.65	11.10	11.94
<i>GDP Growth</i>	1340	9.83	4.82	3.89	7.60	9.30	12.50	16.00
<i>Population Growth</i>	1340	5.47	5.56	-2.77	2.23	5.30	8.49	14.74
City Firm Characteristics (Asset-weighted averages by headquarters city)								
<i>Turnover</i>	1340	0.72	0.33	0.34	0.53	0.68	0.84	1.24
<i>ROA</i>	1340	7.56	8.71	-0.95	4.59	7.40	10.17	15.59
<i>Leverage</i>	1340	0.50	0.21	0.27	0.40	0.47	0.56	0.79
<i>Size</i>	1340	21.87	0.90	20.43	21.30	21.86	22.43	23.28
<i>Cash</i>	1340	0.06	0.06	-0.02	0.03	0.05	0.08	0.14
<i>Capex</i>	1340	0.06	0.04	0.01	0.04	0.05	0.07	0.12
<i>Profit Margin</i>	1340	0.07	0.11	-0.10	0.04	0.09	0.12	0.18
<i>Stock Return</i>	1340	0.17	0.55	-0.36	-0.16	0.01	0.32	1.30

Table 2**AQI as a Determinant of Green Subsidies**

This table presents the determinants of policies related to air pollution treatment. In Models (1) to (4), we examine the cross-city determinants of the environmental subsidy in versions of the Fama-MacBeth (1973) cross-sectional specification:

$$Green\ Subsidy_{j,t} = a_0 + a_1 \times AQI_{j,t-1} + A \times X_{j,t-1} + \varepsilon_{j,t},$$

where $Green\ Subsidy_{j,t}$ denotes the log RMB level of the total environmental subsidy level for city j in year t , $AQI_{j,t-1}$ is the average Air Quality Index of the previous year, and the vector $X_{j,t-1}$ is a set of control variables, including city characteristics and firm characteristics averaged at the city level (weighted according to the value of firm assets). In Models (5) and (6), the city's log RMB level of $Waste\ Gas\ Treatment_{j,t}$ expenditure is the dependent variable. City characteristics include log consumption, log GDP, GDP growth, and population growth. Firm characteristics include asset-weighted averages of the city's firms' asset turnover ratio, return on assets, leverage ratio, firm size, cash holdings, capital expenditures, profit margin, and stock return. The sample period is $t = 2003$ to 2019 and includes up to 352 cities. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	<i>Green Subsidy</i>				<i>Waste Gas Treatment</i>	
<i>AQI</i>		0.05***		0.05***	0.04***	0.04***
		(4.88)		(3.34)	(5.08)	(4.93)
<i>Consumption</i>	-0.31	0.03	0.27	0.65	-1.40	-1.03
	(-0.63)	(0.06)	(0.50)	(1.17)	(-1.29)	(-1.08)
<i>GDP</i>	2.97***	2.93***	2.56***	2.54***	2.37**	2.22**
	(5.67)	(5.69)	(5.52)	(5.43)	(2.58)	(2.49)
<i>GDP Growth</i>	-0.02	-0.02	-0.01	-0.00	0.04	0.05
	(-0.65)	(-0.62)	(-0.23)	(-0.03)	(1.09)	(1.23)
<i>Population Growth</i>	-0.01	0.02	0.01	0.05	-0.11***	-0.10***
	(-0.29)	(0.86)	(0.44)	(1.43)	(-3.77)	(-3.68)
Average firm characteristics	NO	NO	YES	YES	NO	YES
Avg R ²	0.15	0.18	0.24	0.26	0.22	0.33
Observations	1825	1825	1813	1825	1825	1813

Table 3**Influence of Weather on the Air Quality Index (AQI)**

This table estimates the influence of wind and rain on AQI in both heating and non-heating seasons and in northern cities, with colder climates, separately. The specification is:

$$AQI_{j,t,s} = b_0 + b_1 \times Wind\ Speed_{j,t,s} + b_2 \times Rain\ Volume_{j,t,s} + B \times X_{j,t} + \delta_j + \delta_t + \varepsilon_{j,t,s},$$

where $AQI_{j,t,s}$ denotes the average daily AQI of city j in the heating or the non-heating season s of year t , $Wind\ Speed_{j,t,s}$ is the average daily wind speed in that period, and $Rain\ Volume_{j,t,s}$ is the average daily rain volume in that period. The vector $X_{j,t}$ is a set of city characteristics including log consumption, log GDP, GDP growth rate, and population growth rate. Models (1) and (2) compare the influence of wind speed on AQI between heating season and non-heating season in the sample of all cities. Models (3) and (4) compare the influence of wind speed on AQI between the heating season and non-heating season in cities north of the Qinling Mountain-Huai River line. The heating season is November through March. The non-heating season is from April through October. The sample period is $t = 2003$ to 2018 and includes up to 352 cities. Standard errors are clustered by city and year. Robust t -statistics are reported in parentheses. *, **, and *** indicate significance levels of 10%, 5%, and 1%, respectively.

	(1)	(2)	(3)	(4)
	<i>AQI</i>			
	All Cities		Northern Cities	
	Heating	Non-heating	Heating	Non-heating
<i>Wind Speed</i>	-3.15*** (-3.52)	-2.63*** (-3.48)	-5.13*** (-3.09)	-2.34* (-1.78)
<i>Rain Volume</i>	-2.62*** (-4.82)	-1.73*** (-4.62)	-1.60 (-0.49)	-3.33** (-2.41)
<i>Consumption</i>	10.19** (2.20)	-0.68 (-0.16)	-22.44** (-2.04)	-13.87 (-1.40)
<i>GDP</i>	-2.34 (-1.36)	-1.78 (-1.40)	-1.09 (-0.40)	-0.42 (-0.24)
<i>GDP Growth</i>	-0.09 (-1.18)	0.00 (0.04)	0.45** (2.26)	0.41** (2.27)
<i>Population Growth</i>	0.08 (0.71)	0.08 (0.79)	0.49*** (2.68)	0.17 (0.94)
City and year fixed effects	YES	YES	YES	YES
City and year clustering	YES	YES	YES	YES
Adj R ²	0.98	0.98	0.98	0.97
Observations	1699	1699	845	845

Table 4**Policy Uncertainty and Green R&D**

This table presents the impact of environmental subsidy policy uncertainty on green R&D investment by firms in a city. Model (1) presents the OLS regression result and Model (2) to Model (5) report the 2SLS results. In the first stage, we estimate:

$$PU_{j,t-5:t} = c_0 + c_1 \times SD \text{ Windy Days}_{j,t-6:t-1} + c_2 \times SD \text{ Rainy Days}_{j,t-6:t-1} + c_3 \times AQI_{j,t} + C \times X_{j,t} + \delta_j + \delta_t + \varepsilon_{j,t},$$

where the dependent variable $PU_{j,t}$ denotes the policy uncertainty of city j , measured as the standard deviation of the residual of the cross-sectional regression, $Green \text{ Subsidy}_{j,t} = a_0 + A \times X_{j,t-1} + \varepsilon_{j,t}$ (the specification in Model (1) of Table 2) in the six-year period ending year t . $SD \text{ Windy Days}_{j,t-6:t-1}$ and $SD \text{ Rainy Days}_{j,t-6:t-1}$ are the standard deviations of the number of windy days and rainy days per year, respectively, in the six-year period ending year $t - 1$. The vector $X_{j,t}$ is a set of control variables, including AQI as well as city and average firm characteristics. We then use the component of policy uncertainty due to weather variability,

$$\widehat{PU}_{j,t-5:t} = \hat{c}_1 \times SD \text{ Windy Days}_{j,t-6:t-1} + \hat{c}_2 \times SD \text{ Rainy Days}_{j,t-6:t-1},$$

to explain green R&D investment in the second stage:

$$Green \text{ R\&D}_{j,t+1} = d_0 + d_1 \times \widehat{PU}_{j,t-5:t} + d_2 \times Avg \text{ Green Subsidy}_{j,t-5} + D \times X_{j,t} + \delta_j + \delta_t + \eta_{j,t+1},$$

where $Green \text{ R\&D}$ is the log of city aggregate environmental R&D, $Avg \text{ Green Subsidy}$ is the city's characteristics-adjusted environmental subsidy over the prior six years, and other variables are as above. The sample period is generally $t = 2009$ to 2019 . All specifications include city and year fixed effects. Standard errors are clustered by city and year. Robust t -statistics are reported in parentheses. *, **, and *** indicate 10%, 5%, and 1% levels of significance, respectively.

	(1)	(2)	(3)	(4)	(5)
	<i>Green R&D</i>	<i>PU</i>		<i>Green R&D</i>	
	OLS regression	1 st Stage		2 nd Stage	
$\widehat{PU}$ (or <i>PU for OLS</i>)	-0.29** (-2.02)		-2.39*** (-2.82)		-2.23*** (-2.64)
<i>Avg Green Subsidy</i>				0.36*** (2.91)	0.34*** (2.76)
<i>SD Windy Days</i>		0.01*** (3.27)			
<i>SD Rainy Days</i>		0.16*** (3.71)			
<i>AQI</i>	0.01 (0.44)	-0.00 (-1.10)	0.02 (1.45)	0.03** (2.02)	0.02 (1.41)
<i>Consumption</i>	0.38 (0.11)	-0.07 (-0.08)	-0.97 (-0.34)	-0.83 (-0.30)	-1.17 (-0.41)
<i>GDP</i>	1.32* (1.89)	-0.56*** (-2.76)	0.15 (0.16)	1.64** (2.20)	0.28 (0.32)
<i>GDP Growth</i>	0.05 (1.50)	0.01 (0.50)	0.01 (0.29)	0.00 (0.05)	0.01 (0.19)
<i>Population Growth</i>	0.04 (0.64)	-0.05*** (-3.91)	-0.08 (-1.11)	0.04 (0.67)	-0.07 (-0.98)
City and firm control variables	YES	YES	YES	YES	YES
City and year fixed effects	YES	YES	YES	YES	YES

City and year clustering	YES	YES	YES	YES	YES
1 st Stage F		14.5			
Adj R2	0.84	0.89	0.84	0.84	0.84
Observations	1340	1340	1340	1340	1340

Table 5**Policy Uncertainty and Green R&D: Robustness**

This table presents robustness checks to the results in Table 4. The dependent variable in the first-stage regressions is $PU_{j,t}$, leading to the construction of $\widehat{PU}_{j,t-5:t}$, and the dependent variable in the second-stage regressions is $Green\ R\&D_{j,t+1}$. The variations are as follows. Model (1) excludes city observations with no reported environmental R&D. Model (2) uses two-year average AQI and Model (3) adds the six-year rolling standard deviation of weather-adjusted AQI as control variables in both stages. Weather-adjusted AQI is obtained by estimating $AQI_{j,t} = \alpha + b_1 \times Windy\ Days_{j,t} + b_2 \times Rainy\ Days_{j,t} + b_3 \times X_{j,t} + \varepsilon_{j,t}$, and then removing the weather components from AQI to yield $Adj\ AQI_{j,t} = AQI_{j,t} - b_1 \times Windy\ Days_{j,t} - b_2 \times Rainy\ Days_{j,t}$. Model (4) uses Model (3) of Table 2 (rather than Model (1) of Table 2) to estimate a subsidy uncertainty variable that controls for both city and average firm characteristics. Model (5) shows the second-stage results if policy uncertainty is based on subsidies starting in 2007 (resulting in the second-stage observations beginning in $t = 2013$). Model (6) shows a placebo test using Total R&D net of Green R&D as the dependent variable. Model (7) shows a placebo test in which we use Capital Expenditure as the dependent variable. Model (8) shows a placebo test using Capital Expenditure of Real Estate Firms as the dependent variable. $Avg\ Green\ Subsidy_{j,t-5:t}$ is the six-year moving average of the characteristics-adjusted environmental subsidy. The sample period is $t = 2009$ to 2019 . All specifications include city and year fixed effects, and standard errors are clustered by city and year. Robust t -statistics are in parentheses. *, **, and *** indicate 10%, 5%, and 1% levels of statistical significance, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	<i>Green R&D</i>					<i>Placebo Dependent Variables</i>		
	Excluding city obs with Green R&D = 0	Two-year Avg AQI	AQI and SD Adj AQI	Subsidy controlling for average firm chars	Subsample from 2013	Y = Total R&D - Green R&D	Y = Capital Expenditure	Y = Capital expenditure of real- estate firms
$\widehat{PU}$	-2.98** (-2.43)	-2.23*** (-2.59)	-2.22*** (-2.63)	-2.25*** (-2.69)	-5.14*** (-3.17)	0.83 (1.48)	-0.00 (-0.44)	0.03 (0.32)
<i>Avg Green Subsidy</i>	0.11 (0.75)	0.34*** (2.78)	0.34*** (2.76)	0.27** (2.10)	0.00 (0.01)	0.01 (0.11)	-0.00 (-0.11)	0.02** (2.06)
City and aggregated firm- level characteristics	YES	YES	YES	YES	YES	YES	YES	YES
City and year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
City and year clustering	YES	YES	YES	YES	YES	YES	YES	YES
Adj R ²	0.85	0.84	0.84	0.84	0.88	0.96	0.86	0.96
Observations	1127	1340	1340	1340	1039	1340	1340	1340

Table 6**Policy Uncertainty and Green R&D: Firm Differences**

This table presents the effect of environmental subsidy policy uncertainty on firm-level green R&D. We estimate:

$$\text{Green R\&D}_{i \in j, t+1} = e_0 + e_1 \times \widehat{PU}_{j, t-5:t} + e_2 \times \text{Firm Type}_{i \in j, t} \times \widehat{PU}_{j, t-1} + e_3 \times \text{Avg Green Subsidy}_{j, t-5:t} + E \times X_{i \in j, t} + \delta_i + \delta_t + \eta_{i \in j, t+1},$$

where $\text{Green R\&D}_{i \in j, t+1}$ is the logarithm of the environmental R&D investment made by firm i located in city j . The construction of city-level weather-driven policy uncertainty $\widehat{PU}_{j, t-5:t}$ follows Table 4. $\text{Firm Type}_{i \in j, t}$ represents indicator variables for firm characteristics: green tech, metropolitan area headquarters, large size (top assets decile), state-owned enterprises, high (above-median) reliance on environmental subsidies, registration and office addresses located in different cities, manufacturing industry, chemical industry, and public facilities. The CSRC Industry Classification Code is in parentheses. $\text{Avg Green Subsidy}_{j, t-5:t}$ is the six-year moving average of the characteristics-adjusted environmental subsidy. The sample period is $t = 2009$ to 2019 . All specifications include city characteristics, firm characteristics, and firm and year fixed effects. Robust t -statistics are reported in parentheses. *, **, and *** indicate 10%, 5%, and 1% levels of statistical significance, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	<i>Green R&D (Firm)</i>											
$\widehat{PU}$	-0.40*** (-2.72)	-0.17 (-1.09)	-0.38** (-2.43)	-0.41*** (-2.75)	-0.39** (-2.53)	-0.18 (-1.19)	-0.40*** (-2.61)	0.10 (0.63)	-0.32** (-2.11)	-0.37** (-2.50)	-0.40*** (-2.69)	0.58*** (3.31)
$\widehat{PU} * \text{Green Tech}$		-0.58*** (-7.47)										-0.48*** (-6.02)
$\widehat{PU} * \text{Metropolitan}$			-0.07 (-0.73)									-0.22** (-2.41)
$\widehat{PU} * \text{Large}$				0.11 (0.71)								-0.08 (-0.60)
$\widehat{PU} * \text{SOE}$					-0.04 (-0.46)							-0.07 (-0.85)
$\widehat{PU} * \text{Subsidy Reliant}$						-0.62*** (-7.77)						-0.35*** (-4.20)
$\widehat{PU} * \text{Different Cities}$							-0.01 (-0.10)					-0.10 (-0.99)
$\widehat{PU} * \text{Manufacturing (C)}$								-0.78*** (-10.13)				-0.66*** (-7.70)
$\widehat{PU} * \text{Chemical (C25, C26, C28, C29, C41, C43)}$									-0.76*** (-5.07)			-0.41*** (-2.59)

$\widehat{PU}$ * Metal (C31, C32, C65, C67, C33)										-0.92***		-0.51**
										(-4.44)		(-2.39)
$\widehat{PU}$ * Public Facility (N)											-0.91**	-1.12***
											(-2.52)	(-3.02)
Avg Green Subsidy	0.06*	0.05	0.06*	0.06*	0.06*	0.06*	0.06*	0.06*	0.05*	0.06*	0.06*	0.05
	(1.74)	(1.56)	(1.74)	(1.78)	(1.76)	(1.72)	(1.77)	(1.87)	(1.67)	(1.75)	(1.70)	(1.48)
City and firm characteristics	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Firm and year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Adj R2	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43
Observations	26189	26189	26189	26189	26189	26189	26189	26189	26189	26189	26189	26189

Table 7**Policy Uncertainty and Green R&D: Subsidy Reliance and Financing**

This table presents the effect of environmental subsidy policy uncertainty on firm-level green R&D within green tech firms. We estimate the same specifications as in Model (2) of Table 6, but with double interactions. $Green\ R\&D_{i \in j, t+1}$ is the logarithm of the environmental R&D investments made by firm i located in city j . The construction of city-level weather-driven policy uncertainty $\widehat{PU}_{j, t-5:t}$ follows earlier tables. $Green\ Tech_{i \in j, t}$ represents green tech firms. *Subsidy Reliant* is an indicator for high (above-median) reliance on environmental subsidies. *Over-indebtedness* is CSMAR's calculation of the actual debt ratio minus the target debt ratio. *Interest Coverage Ratio A* is CSMAR's calculation of Net Profit plus Income Tax Expense plus Interest Expense, all divided by Interest Expense. *Interest Coverage Ratio B* excludes Income Tax Expense. $Avg\ Green\ Subsidy_{j, t-5:t}$ is the six-year moving average of the characteristics-adjusted environmental subsidy. The sample period is $t = 2009$ to 2019 . All specifications include city characteristics, firm characteristics, additional double interaction terms, and firm and year fixed effects. Robust t -statistics are reported in parentheses. *, **, and *** indicate 10%, 5%, and 1% levels of statistical significance, respectively.

	(1)	(2)	(3)	(4)
	<i>Green R&D (Firm)</i>			
$\widehat{PU}$	-0.04 (-0.24)	-0.21 (-1.28)	-0.19 (-1.21)	-0.19 (-1.21)
$\widehat{PU} * Green\ Tech$	-0.39*** (-3.95)	-0.60*** (-7.11)	-0.57*** (-6.99)	-0.57*** (-6.99)
$\widehat{PU} * Green\ Tech * Subsidy\ Reliant$	-0.33** (-2.02)			
$\widehat{PU} * Green\ Tech * Over-indebtedness$		-0.55** (-2.09)		
$\widehat{PU} * Green\ Tech * Interest\ Coverage\ Ratio\ A$			0.30* (1.81)	
$\widehat{PU} * Green\ Tech * Interest\ Coverage\ Ratio\ B$				0.33* (1.91)
<i>Avg Green Subsidy</i>	0.05 (1.54)	0.06 (1.63)	0.06 (1.62)	0.06 (1.62)
Double interactions and firm characteristics	YES	YES	YES	YES
City and firm characteristics	YES	YES	YES	YES
Firm and year fixed effects	YES	YES	YES	YES
Adj R2	0.43	0.43	0.43	0.43
Observations	26189	24068	24681	24681

Table 8**Policy Uncertainty and Green R&D Employment**

This table presents the impact of subsidy policy uncertainty on the number of environmental R&D researchers across firms in a city. *Green R&D Employment* is the logarithm of one plus the total number of R&D and technical employees among firms with environmental R&D investment headquartered in the city. The construction of weather-driven policy uncertainty $\widehat{PU}_{j,t-5:t}$ follows Table 4. In addition to city and average firm characteristics as controls, *Avg Green Subsidy* $_{j,t-5:t}$ is the six-year moving average of the characteristics-adjusted environmental subsidy. All specifications include city and year fixed effects, and standard errors are clustered by city and year. The sample period is $t = 2009$ to 2019 . Robust t -statistics are reported in parentheses. *, **, and *** indicate 10%, 5%, and 1% levels of statistical significance, respectively.

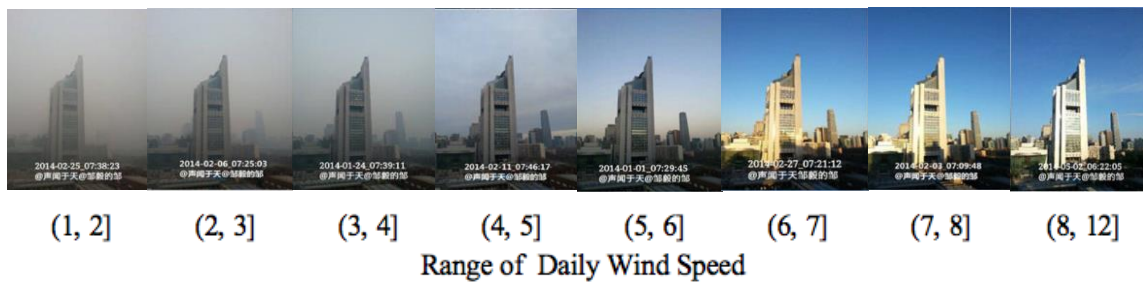
	(1)	(2)	(3)
	<i>Green R&D Employment</i>		
$\widehat{PU}$	-1.07*** (-3.41)		-1.00*** (-3.22)
<i>Avg Green Subsidy</i>		0.16*** (3.61)	0.15*** (3.42)
City and average firm characteristics	YES	YES	YES
City and year fixed effects	YES	YES	YES
City and year clustering	YES	YES	YES
Adj R2	0.85	0.85	0.85
Observations	1340	1340	1340

Fig 1.

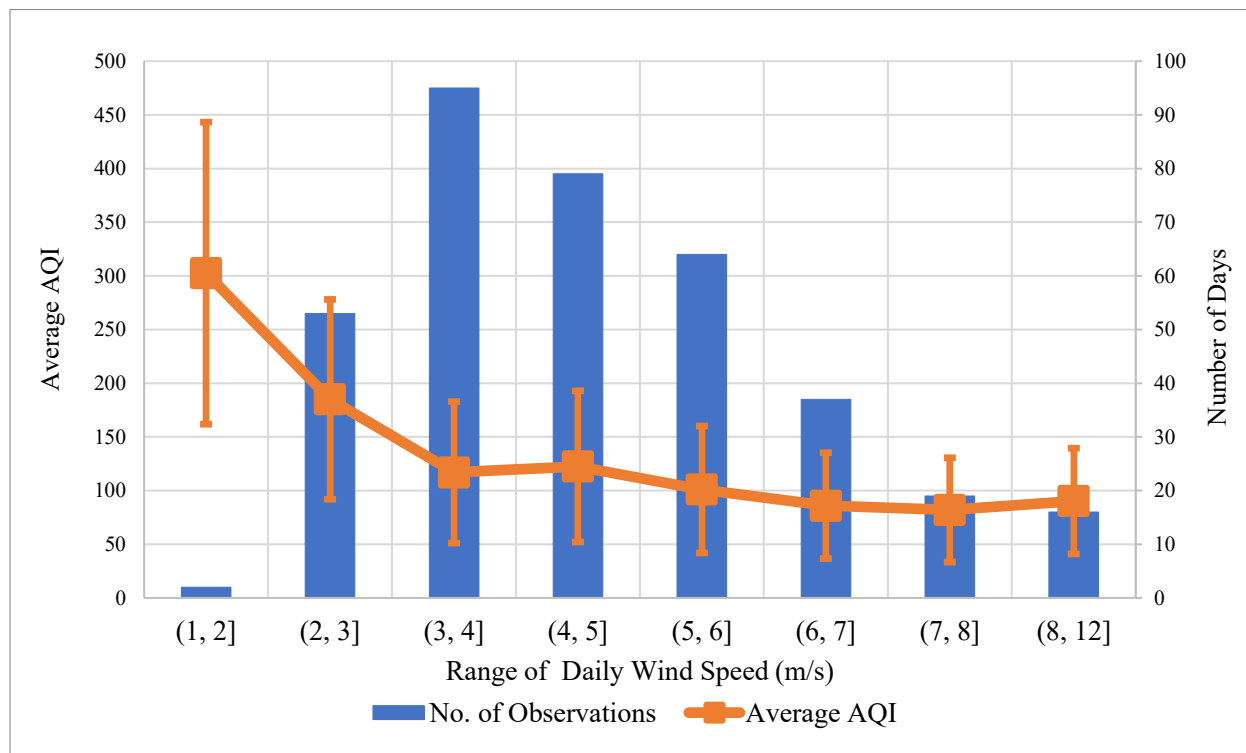
AQI and Weather.

This figure illustrates how air quality depends on wind and rain. Panel A shows pictures of the same building in Beijing taken on different days in 2014. Source: <https://tinyurl.com/4d5kcjk4>. From left to right, the wind speed is 1-2, 2-3, etc. meters per second. Panel B plots the corresponding average AQI in Beijing in 2014 as a function of average wind speed and the number of days in 2014 at each wind speed. Rain is scarce in Beijing, so Panel C pools 2013-2018 data for all cities in the sample to show the effect of rain on AQI. The nonzero daily rain levels tracked by the China Meteorological Association are: < 10mm, 10-25mm, 25-50mm, 50-100mm, 100-200mm, and > 200mm.

Panel A. Photos from Beijing Versus Wind Speed, 2014



Panel B. AQI and Wind Speed in Beijing, 2014



Panel C. AQI and Rain Volume, all cities, 2013-2018

